

Nextcloud & WireGuard

On a Home Ubuntu Server

Design and implementation of a self-hosted NAS with end-to-end encrypted remote access, conceived as a small-scale system architecture case study.

Context & motivation

- Personal data sovereignty
- Zero marginal cost : using existing hardware
- A hands-on learning environment
- Remote access

Technical stack

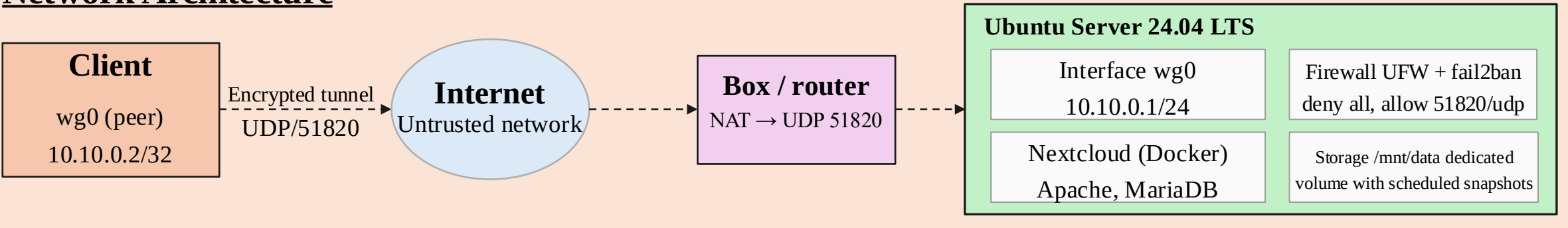
Ubuntu 24.04 LTS, Docker, Nextcloud, WireGuard, MariaDB, UFW, fail2ban, ZFS/RAID

Summary

This project documents the setup of a personal server running Nextcloud as a software-based NAS (files, calendar, contacts, and photos), combined with a WireGuard tunnel for secure remote access, without directly exposing the server to the Internet.

Métrique	Observed value	Target
Tunnel latency (LAN → server)	≈ 3–5 ms	< 10 ms
Nextcloud synchronization throughput	To be measured	≥ 50 Mo/s
WireGuard handshake time	< 1 s	—
Publicly exposed ports	1 (UDP)	1

Network Architecture



The client never directly accesses the application services: all remote traffic passes through the WireGuard tunnel before being processed by the firewall and then forwarded to Nextcloud.